

FusionFinance

Quantitative discipline meets agentic investigation – with evidence and risk gates between language and capital.

SUBMISSION THESIS

ML proposes. Agents investigate. The verifier challenges. The risk engine decides.

Autonomous trading is split between two incomplete systems

Pure ML is disciplined but context-blind

Structured models rank opportunities consistently, but do not understand a guidance change, supply shock, or accounting caveat.

Pure LLM is contextual but poorly calibrated

Agents synthesize narrative, but can hallucinate evidence, overstate confidence, and translate prose into unstable position sizes.

Neither system should control a portfolio without the other system checking its blind spots.

Three arms belong under one contract; today's replay does not yet meet it

Pure ML

Structured data only. No agent interpretation.

Pure LLM

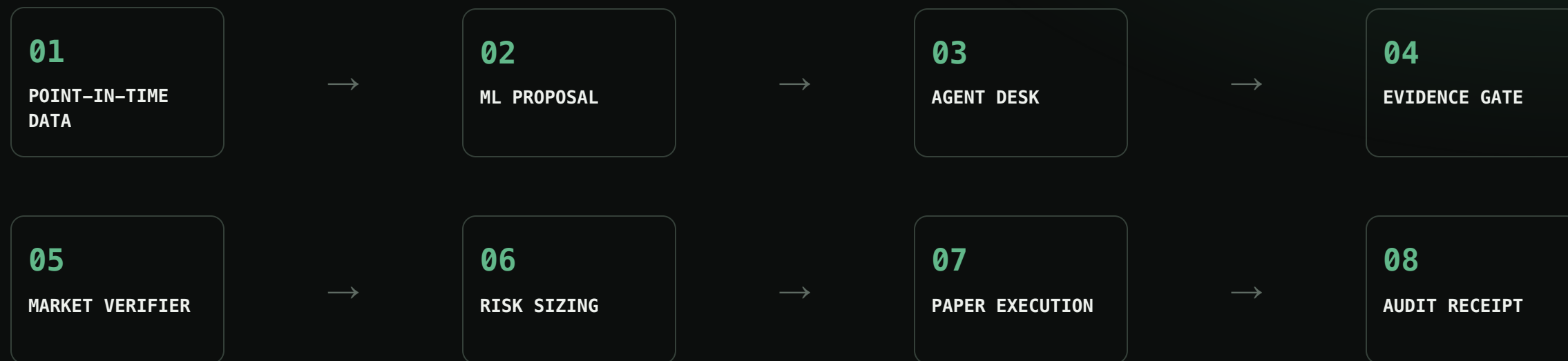
Qualitative interpretation only. No proprietary ML proposal.

FusionFinance

ML proposal, investigation, evidence challenge, and risk sizing.

- Controlled destination: same universe, capital, timestamps, costs, slippage, leverage, limits, kernel, and rebalance cadence.
- Current legacy replay: same window and timestamps, but different legacy kernels and no shared cost model.
- The next shared-kernel run is the falsifiable comparison; the current snapshot demonstrates the product and metric path.

Each component receives a job it can actually perform



Recorded AMD-backed stage: a separate walk-forward model-training workload using PyTorch on ROCm/HIP. It did not generate the displayed legacy curves.

Missing contract fields force a visible rejection

MU • Sell proposed

Recorded confidence: 0.60. Expected outcome: not recorded. Rationale: parabolic 3m surge with volume blow-off.

Final action • Reject • Position 0

The legacy record has no horizon, evidence sources, prospective seal, or complete provenance.

This projection is not proof that the full pipeline executed historically. It demonstrates that the modern gate preserves missing fields and blocks an incomplete candidate from capital.

FAIL horizon and expected outcome
FAIL evidence references
FAIL prospective seal
FAIL complete provenance
RESULT evidence_contract_reject

The strongest stored line is not a controlled win

ARM	TOTAL RETURN	SHARPE	SORTINO	MAX DRAWDOWN	VOLATILITY
Pure ML	+1.7%	+0.76	+1.03	-2.9%	5.4%
Pure LLM	-10.3%	-0.90	-1.28	-23.8%	24.5%
FusionFinance legacy line	+10.1%	+2.05	+3.48	-3.9%	11.1%
SPY benchmark	+8.7%	+1.39	+2.10	-8.9%	14.6%

CLAIM BOUNDARY

Fusion has sparse five-session marks: 21 nonzero changes across 109 stored values, weakening drawdown comparability. The legacy line contains the ML champion and every historical LLM thesis fails today's evidence contract. This is not causal proof of outperformance.

Fusion is a sequence of challenges, not an average of opinions

ML proposes

Expected return, uncertainty, volatility, and liquidity define the candidate set.

Agents investigate

Market, news, fundamentals, and risk roles search for context and falsifiers in parallel.

The system decides

Evidence can reject. The independent market head can veto. Portfolio risk controls size what survives.

The LLM never gets a direct path to execution, and the model never gets to ignore contradictory context.

A self-reported receipt chain records an AMD gfx1100 training workload

231.29 s

recorded walk-forward GPU training time

51.5 GB

VRAM bytes reported by rocm-smi

```
MATCH AMD/ATI vendor field
MATCH gfx1100 architecture field
MATCH ROCm/HIP 7.2 runtime
MATCH 72 ranker retrains
MATCH source receipt SHA-256
NOTE device-name field was blank
```

Cross-checks derive the published values from preserved receipts and hashes protect byte integrity; this is not independent attestation. Exact commercial SKU and utilization are unavailable. The separate workload did not generate the displayed legacy curves.

An auditable decision system matters even when a trade is rejected

- Quantitative research and replay.
- Paper-trading decision support.
- Portfolio monitoring and risk review.
- Human-in-the-loop institutional workflows.
- Point-in-time filing and market features.
- Hash-bound evidence and GPU receipts.
- Explicit abstain and research-only states.
- Documented leakage, survivorship, cost, and sample limitations.

RESEARCH SOFTWARE, NOT A PROFIT PROMISE

Backtests can overfit. Historical LLM analysis can leak through pretraining. Paper fills differ from live execution. FusionFinance makes those constraints visible and machine-checkable.

Not the agent. Not the model. The auditable fusion.

Autonomous finance should not be controlled by an unverified language model or an isolated statistical model. FusionFinance combines machine learning, agentic reasoning, quantitative verification, and risk control into one auditable decision system.

Repository: <https://github.com/FusionCube18712/FusionFinance>

Live demo: <https://fusionfinance2.vercel.app> · Replay artifact: [results/demo_run.json](#)